Hidden Path Selection Network for Semantic Segmentation of Remote Sensing Images

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***Abstract*— Targeting at depicting land covers with pixelwise semantic categories, semantic segmentation in remote sensing images needs to portray diverse distributions over vast geographical locations, which is difficult to be achieved by the homogeneous pixelwise forward paths in the architectures of existing deep models. Although specific algorithms have been designed to select pixelwise adaptive forward paths for natural image analysis, it still lacks theoretical supports on how to obtain optimal selections. In this article, we provide mathematical analyses in terms of the parameter optimization, which guides us to design a method called hidden path selection network (HPS-Net). With the help of hidden variables deriving from an extra mini-branch, HPS-Net is able to tackle the inherent problem about inaccessible global optimums by adjusting the direct relationships between feature maps and pixelwise path selections in existing algorithms, which we call hidden path selection. For the better training and evaluation, we further refine and expand the 5-class Gaofen image dataset (GID-5) to a new one with 15 land-cover categories, i.e., GID-15. The experimental results on both GID-5 and GID-15 demonstrate that the proposed modules can stably improve the performance of different deep structures, which validates the proposed mathematical analyses.**

***Index Terms*— Benchmark dataset, hidden path selection, remote sensing image, semantic segmentation.**

I. Introduction

SEMANTIC segmentation in remote sensing images [1], [2], [3], [4], which targets at identifying land covers with pixelwise semantic categories, is of considerable significance among many land use management-related tasks [5], [6], [7]. Due to the complexity of real-world environments, tackling issues such as the diversity of land-cover distributions over different geographical locations is an important topic among existing investigations that are related to remote sensing image interpretation [8], [9], [10], [11], [12], [13], [14]. As a consequence, studying semantic segmentation methods to depict diverse complicated land-cover distributions simultaneously has drawn increasing attention in the remote sensing community [15], [16], [17].

For implementing semantic segmentation, the methods to transfer the spectral signals into digital features first came into researchers' views, and classical classifiers in machine learning [18], [19] could be applied to predict pixelwise land-cover categories [17], [20], [21], where elaborate computation procedures could be exploited to obtain features corresponding to diverse land-cover distributions. Nevertheless, with predefined man-made calculations, hand-crafted features are hard to represent numberless different land-cover distributions in real-world environments, which has encouraged following researches to focus on improving the flexibility and representation capabilities of features through learnable deep structures [3], [22], [23].

However, although optimal parameters could be adapted to land-cover distributions existing in training data samples, the deep structures are often static after being learned in the training processes, which could lose the adaptability to complicated real-world environments during the inference procedures [24]. To address this issue, many researchers have designed various elaborate structures to capture complex contextual information [25], [26], [27] not only for remote sensing images but also for natural ones.

Subsequently, neural architecture search (NAS) [28] is proposed to develop models with automatically acquired structures to adapt to diverse data distributions. Furthermore, some researchers have proposed structures, called dynamic routing, to select forward paths adaptive to each data sample through designed activating factors [24] instead of adapting to data distributions of the whole dataset as in [28]. However, with all pixels sharing the same activating factors, such methods might ignore the diversity over different image areas, the effect of which is remarkable in remote sensing images due to the vast covered geographical areas with diverse land-cover distributions. Consequently, the gated path selection network (GPSNet) [29] has been proposed to select pixelwise adaptive forward paths, simply shown at the top of Fig. 1, through designed soft masks instead of same activating factors shared by all pixels, as in [24]. However, considering all items in GPSNet as the variables, the direct functional relationships between soft masks (executing pixelwise path selections through multiplication) and corresponding feature maps would

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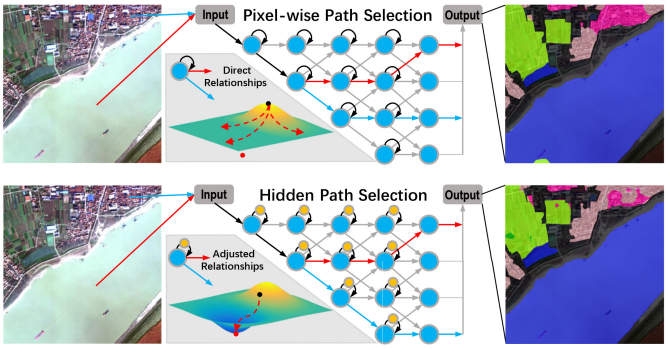


Fig. 1. In this article, we provide mathematical analyses of the parameter optimization in semantic segmentation of remote sensing images, which imply that direct relationships between feature maps and pixelwise path selections in existing algorithms could make global optimums inaccessible. Consequently, we design the hidden path selection to adjust the aforementioned deleterious direct relationships through hidden variables and make the optimal path selections more possible to obtain. Taking all items in deep structures as variables, the manifold for parameter optimization of HPS-Net is more likely to obtain the global optimums compared with GPSNet, simply shown at the bottom-left of each network. Straight lines with different colors represent selected paths for different pixels, while the gray lines indicate the path is not emphasized. Blue circles indicate the feature maps and orange circles indicate the hidden variables.

make the parameter optimization constrained on a narrow high-dimensional manifold instead of the full space, which might exclude the global optimums as illustrated in Fig. 1. In other words, the direct relationships between feature maps and path selections could make it difficult to obtain optimal soft masks, which means the optimal pixelwise forward path selections adaptive to different land-cover distributions are hard to obtain. In summary, how to depict the land-cover distributions in complicated real-world environments for different pixels during the inference procedure has not been solved well by existing algorithms, which is of great interest to investigate.

In this article, we portray the complicated pixelwise land-cover distributions by imitating the hidden Markov chain. As illustrated in Fig. 2, we propose a method, called hidden path selection network (HPS-Net), with independent parallel structures (i.e., a main branch for semantic segmentation and a mini-branch for supporting) by exploiting a sequence of special modules called hidden path module (HP-Module). Considering all items in the main branch as variables, we notice that the high-dimensional manifold where we execute the parameter optimization could be adjusted through utilizing hidden variables highly independent with main branch. Thus, we utilize the HP-Module to obtain a sequence of soft masks by leveraging hidden variables from the mini-branch to select pixelwise adaptive forward paths in the main branch through multiplication, where the adjusted high-dimensional manifold would have higher possibility to contain the global optimums.

Further inspired by the mathematical analyses in terms of parameter optimization, we cut off the gradient flows from soft masks to the main deep structures during feature integration in the HP-Module to improve the training efficiency, while the gradient flows are retained intactly during the multiplication. Then, according to the Taylor expansion with respect to soft masks in HPS-Net, whose values are defined within a small range, the loss function can approximate a convex function to some extent if global optimums are contained in the aforementioned high-dimensional manifold. As a consequence, the usage of nearly freely valuing hidden variables would make soft masks adjusted freely and thus the convexity could help to obtain optimal soft masks (namely optimal pixelwise path selections). As we shall see, with the proposed modules applied on existing deep structures, HPS-Net can better portray the pixelwise land-cover distributions during the inference procedure.

Another important aspect related to semantic segmentation in remote sensing images is the accessibility of well-annotated dataset. Datasets simultaneously possessing high spatial resolution, fine annotations, large quantities, and abundant land-cover categories are hard to acquire and not sufficient enough yet. Meanwhile, the urge for well-annotated large-scale datasets containing images of high spatial resolution is increasing [30], [31]. Although many existing datasets, e.g., [32], [33], [34], [35], contain increasingly abundant land covers to compose different distributions for the researches, the included scenes cannot perfectly reflect the complex real-world environments due to the limited data sample quantities. Recently, a few large-scale datasets, e.g., [36], [37], have been built, but the involved land-cover categories are often insufficient yet and hardly meet the practical demands well. Besides, the practical usage of some other large-scale datasets [38], [39] with numbers of land-cover categories are constrained by the annotation form and spatial resolution, respectively. In this article, we expand the land-cover classification dataset in [3] by refining the five involved land-cover categories in the whole GID-5 [36] into 15 categories, which can better serve the practical demands.

Our main contributions in this article are threefold.

1) We propose a new method, i.e., HPS-Net, to imitate hidden markov chain, which depicts land-cover distributions and selects adaptive forward paths for each pixel through an extra mini-branch highly independent with arbitrary main deep structures.
2) We provide mathematical analyses on the parameter optimization with taking all items in deep structures as variables to explain the inaccessibility of global optimums, guided by which we design detailed structures of the HPS-Net and propose the HP-Module to improve the possibility to obtain the global optimums.
3) We expand the large-scale remote sensing semantic segmentation dataset, i.e., GID-5, into 15 land-cover categories and propose the new GID-15 dataset, which provides a challenging large-scale benchmark platform with abundant land-cover categories and better satisfies the real-world scenarios.

II. Related Work

A. Semantic Segmentation

Early researches in remote sensing semantic segmentation often leveraged and developed the digital features, among which some works [17], [20], [21] rely on the physical

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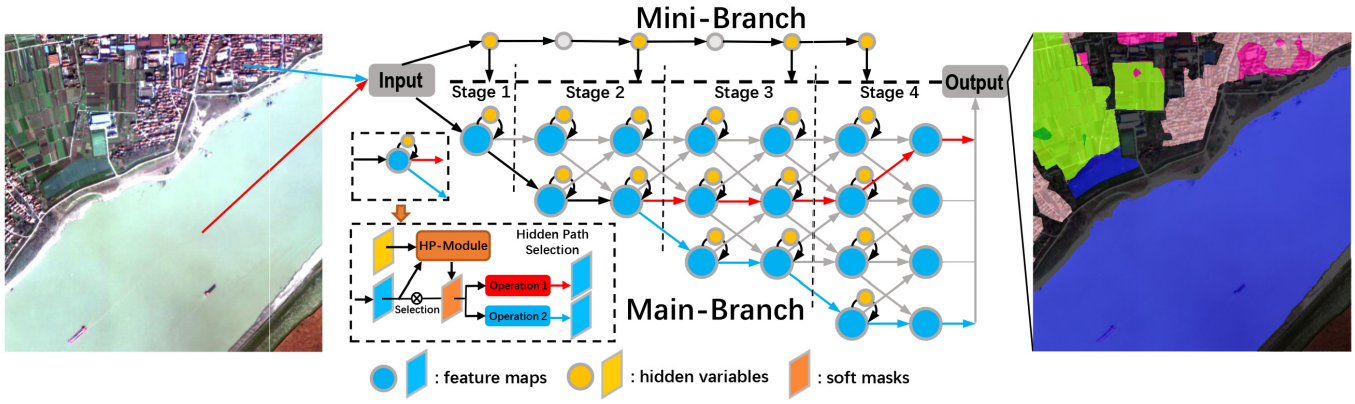


Fig. 2. HPS-Net for semantic segmentation in remote sensing images. HPS-Net utilizes an extra mini-branch to generate hidden variables for the main branch, which can be utilized to obtain soft masks for selecting pixelwise paths by multiplication in the main branch through designed HP-Module. Taking arbitrary deep structures as the main branch, HP-Module integrates feature maps in the main branch and corresponding hidden variables to calculate soft masks, which select paths from existing alternatives (e.g., parallel structures or skip connections) in the structures of main branch (e.g., Resnet) for each pixel. In this article, we take Resnet as the main branch, while the mini-branch utilizes same structures with reduced channel numbers. We divide the mini-branch and main branch into several stages according to the pixel resolution. Last layer of each stage in mini-branch serves as hidden variables for the same stage in main branch. In addition to the existing skip connections in Resnet, the last layer of each stage in main branch would be considered as an extra alternative path for the following stages.

properties of land-covers and some other works [40], [41] capture the local spatial information. Subsequently, many deep structures [42], [43], [44], [45], whose fully connected layers are removed, have been borrowed in semantic segmentation models not only for remote sensing images but also for natural images [46], [47], [48]. Numerous learnable parameters in deep structures can help obtain features with stronger representation capabilities. Especially, researches on depicting different object distributions, such as the existence of multiscale objects, have been hot spots for a long time [25], [26], [27]. Meanwhile, the community were inspired to research on NAS [28] for developing semantic segmentation models with automatically acquired structures instead of artificial structures. The architectural hyperparameters in these automatically acquired structures could make it more adaptive to different data distributions.

Recently, some works have been proposed to focus on developing dynamic structures instead of static architectures (even automatically acquired). Specifically, activating factors obtained by designed soft conditional gates in [24] dynamically emphasize adaptive forward paths during the inference procedures. However, land-cover distributions often vary significantly within one remote sensing image due to the wide range of covered areas. Thus, the diversity of land-cover distributions could require architectures to dynamically select adaptive forward paths for each pixel rather than for the whole image in [24].

In this article, we propose the HPS-Net by imitating the hidden Markov chain, where an extra mini-branch assists to better select pixelwise adaptive forward paths in the main branch for implementing semantic segmentation.

B. Data-Dependent Architectures

Following the idea of [28], several works have been proposed to obtain architectural hyperparameters based on the image contexts among the whole dataset [38], [39], [49]. Especially, Zoph *et al.* [49] firstly proposed cell-level architecture search space, while Zhang *et al.* [38] was proposed to connect the cells and obtain a densely connected architecture search space. Besides, Zhang *et al.* [39] embedded a directed acyclic graph (DAG) into the search space and designed the differentiable searching process. In summary, all these architectures are data-dependent, which can be adaptive to different data distributions.

Furthermore, another kind of data-dependent architecture is the attention mechanism, which has been designed in deep structures [50], [51] to aggregate the spatial attention together with channel attention and then dynamically adjust contextual information for every pixel. Especially, nonlocal operations were proposed to obtain dynamic attention maps through pixelwise similarities [52]. It is worth noticing that [24] is also data-dependent, where the activation factor can be seen as attention weights for the whole image.

Recently, soft masks, which can be seen as the combination of several normalized attention maps, have been proposed [29] to build data-dependent architectures. Specifically, Geng *et al.* [29] dynamically selects adaptive forward paths for every pixel according to the image contexts with the help of soft masks. However, according to the mathematical analyses in terms of parameter optimization we shall provide in this article, the direct functional relationships between the soft masks and corresponding features in existing pixelwise dynamic structures [29] would make it difficult to obtain the global optimums.

Consequently, we design an HP-Module to take the advantage of hidden variables deriving from the extra mini-branch and adjust the aforementioned deleterious direct relationships, which makes HPS-Net more likely to obtain the optimal pixelwise path selections based on image contexts.

C. Benchmark Dataset for Semantic Segmentation

To support developing semantic segmentation algorithms in remote sensing images, benchmark datasets are important to serve as the basis for training and evaluation. Widely used

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datasets [32], e.g., ISPRS Test Project on Urban Classification and 3-D Building Reconstruction, focuses on urban areas in cities such as Vaihingen. Some contests on other topics, such as IEEE GRSS Data Fusion Contest in 2017 and 2018, also provide semantic segmentation datasets [33], [35] covering several cities, while [34] provides data samples from different platforms. Besides, some datasets concentrating on specific categories, such as buildings and roads, are also provided [53]. Further to expand the quantity of data samples, several large-scale datasets have been created [36], [37], which cover larger geographical areas to ensure the diversity of data samples. However, existing datasets hardly give considerations to both the quantity of images and sufficiency of land-cover categories. For instance, although GID-5 [36] contains a large quantity of data samples and covers wide ranges of geographical areas, the limited five land-cover categories are often insufficient for real-world applications.

In this article, we expand the GID-5 by refining five general categories into 15 elaborated categories and create the GID-15 dataset, which can serve as a challenging benchmark and better meet the demands in real-world applications.

III. Methods

A. Mathematical Formulation

Interpreting the image $\mathbf{I}$ of $c \in \mathbb{N}_+$ channels as a mapping function belongs to a special function family $\mathscr{F}$, we have $\mathbf{I}: \Omega \to \mathbb{R}^c$ defined on the image grid Ω:$\{0, 1, \ldots, H-1\} \times \{0, 1, \ldots, W-1\}$, which would return values of c channels by given a coordinate of position $\mathbf{p}$ in Ω. Consequently, the semantic segmentation model can be interpreted as a mapping function that

$$\forall\ \mathbf{I} \in \mathscr{F}, \quad \mathbf{p} \in \Omega, \quad f(\mathbf{I}, \mathbf{p}) = \arg\max_{l_k \in \mathbf{L}} \mathcal{M}_{\mathbf{I}}(\mathbf{p}). \tag{1}$$

$\mathbf{L} = \{l_k\}_{k \in \mathbb{N}_+ \le N}$ is a label set containing N categories and probability map $\mathcal{M}_{\mathbf{I}}$ measures the probability of each category corresponding to each position $\mathbf{p} \in \Omega$.

The structures of conventional deep models often combine various elaborate units, such as convolutions and nonlinear activation functions, to capture complex contextual information adaptive to various distributions. Therefore, considering each layer in existing deep structures, such as [43], as a basic function consisting of different units, the whole structures of $D \in \mathbb{N}_+$ layers can be interpreted as a sequence of nested composite functions that

$$\begin{aligned} &\forall\ \mathbf{F}_1 = \mathbf{I}, \quad d \in \mathbb{N}_+ < D, \quad \mathbf{F}_{d+1} = f_d\left(\{\mathbf{F}_i\}_{i \in \mathbb{N}_+ \le d}\right) \\ &\mathcal{M}_{\mathbf{I}} = \mathbf{F}_D, \quad f(\mathbf{I}, \mathbf{p}) = \arg\max_{l_k \in \mathbf{L}} \mathcal{M}_{\mathbf{I}}(\mathbf{p}) \end{aligned} \tag{2}$$

where f_d represents the basic function in the dth layer and $\mathbf{F}_d$ represents the corresponding feature maps. Considering in the skip connection structures, f_d receives feature maps $\{\mathbf{F}_i\}_{i \in \mathbb{N}_+ \le d}$ in all the previous layers (not all necessarily participate in calculations) and returns the feature maps $\mathbf{F}_{d+1}$ for f_{d+1} in the next layer. Consequently, the first layer in deep structures receives image $\mathbf{I}$ as the input and feature map $\mathbf{F}_D$ in the last layer serves as the probability map $\mathcal{M}_{\mathbf{I}}$ to measure the pixelwise probability of each category. It is worth noticing that models deriving from NAS can also be interpreted in this form.

To better adapt to complicated real-world scenarios, dynamic routing [24] is proposed to select image-dependent forward paths in deep structures. Specifically, we can interpret dynamic routing as

$$\begin{aligned} &\forall\ \mathbf{F}_1 = \mathbf{I}, \quad d \in \mathbb{N}_+ < D, \quad \mathbf{F}_{d+1} = f_d\left(\{\mathbf{F}_i\}_{i \in \mathbb{N}_+ \le d}, \alpha^d\right) \\ &\mathcal{M}_{\mathbf{I}} = \mathbf{F}_D, \quad f(\mathbf{I}, \mathbf{p}) = \arg\max_{l_k \in \mathbf{L}} \mathcal{M}_{\mathbf{I}}(\mathbf{p}) \end{aligned} \tag{3}$$

where $\alpha^d \in \mathbb{R}^s$ represents the activation factor for total $s \in \mathbb{N}_+$ forward paths in the d-th layer. In the implementation, dynamic routing utilizes learnable structures to compose the mapping function $\mathbf{G}_d : \mathbf{I} \to \mathbb{R}^3$ and obtain α^d for selecting forward paths corresponding to up-scale, equivalent-scale, and down-scale. The activation factor would be multiplied with corresponding feature maps to select paths adaptive to different scale distributions among the whole input image.

B. Gated Path Selection

Although dynamic routing mentioned in Section III-A can adapt to scale distributions during the inference procedure, it rarely considers the diversity over different image areas, which is remarkable in remote sensing images due to the wide range of covered geographical areas. Recently proposed GPSNet [29] extends the idea of dynamic routing, where the utilized soft masks can be seen as the activation factor maps to select adaptive forward paths for each pixel. Specifically, we have

$$\begin{aligned} &\forall\ \mathbf{F}_1 = \mathbf{I}, \quad d \in \mathbb{N}_+ < D, \quad \mathbf{F}_{d+1} = f_d\left(\left\{\mathbf{F}_i \odot \mathbf{W}_i^d\right\}_{i \in \mathbb{N}_+ \le d}\right) \\ &\mathcal{M}_{\mathbf{I}} = \mathbf{F}_D, \quad f(\mathbf{I}, \mathbf{p}) = \arg\max_{l_k \in \mathbf{L}} \mathcal{M}_{\mathbf{I}}(\mathbf{p}) \end{aligned} \tag{4}$$

where $\mathbf{W}_i^d$ is the soft mask of the same spatial size with feature map $\mathbf{F}_i$, where the element in each position serves as a probability for pixelwise forward path selections. The symbol $\odot$ represents the elementwise product. It is worth noticing that $\mathbf{W}^d$ in the traditional layers can be seen as tensors full with $\mathbf{1}$. In the implementation, GPSNet directly utilizes $\mathbf{F}_i$ to generate $\mathbf{W}_i^d$, namely $\mathbf{W}_i^d = g_i^d(\mathbf{F}_i)$ highly dependent with $\mathbf{F}_i$ for each i with g_i^d being a differentiable function, which would make it difficult to obtain the global optimums of the problem as we analyze in the following texts.

Noticing that with features in deep structures being considered as variables determined by parameter $\boldsymbol{\theta}_f$ in $\{f_d\}_{d \in \mathbb{N}_+ < D}$, the optimization of $\boldsymbol{\theta}_f$ can be seen as searching the global optimums within a high-dimensional manifold $\mathcal{H}_f$ in the full space, whose shape is determined by the structures we design for $\{f_d\}_{d \in \mathbb{N}_+ < D}$. Moreover, optimizing parameters $\boldsymbol{\theta}_f$ in all $\{f_d\}_{d \in \mathbb{N}_+ < D}$ during the training process would guide us to find $\mathbf{F}^*{}_i$ and $\mathbf{W}_i^{*d}$ for each $\mathbf{I}$ determined by the optimal parameters. Starting from this view, we calculate $\partial L / \partial \mathbf{F}^*{}_i$ and $\partial L / \partial \mathbf{W}*_i^d$ with L as the loss function such that

$$\begin{aligned} &\frac{\partial L}{\partial \mathbf{F}^*{}_i} = \sum_{d < D} \left(\frac{\partial L}{\partial f_d} \cdot \frac{\partial f_d}{\partial \mathbf{F}^*{}_i \odot \mathbf{W}_i^{*d}} \right) \cdot \mathbf{W}_i^{*d} \\ &\forall d < D, \quad \frac{\partial L}{\partial \mathbf{W}_i^{*d}} = \frac{\partial L}{\partial f_d} \cdot \frac{\partial f_d}{\partial \mathbf{F}^*{}_i \odot \mathbf{W}_i^{*d}} \cdot \mathbf{F}^*{}_i \end{aligned} \tag{5}$$

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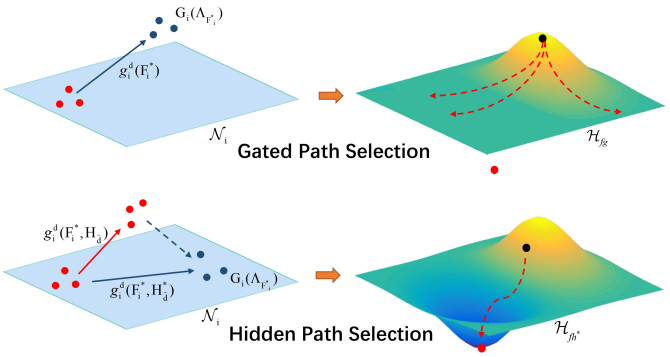


Fig. 3. In gated path selection, g_i^d is hard to meet the required property that numbers of specific points in $\mathcal{N}_i$ should be mapped within $\mathcal{N}_i$ through g_i^d (i.e., G_i), which could exclude the global optimums (red point) outside the manifold $\mathcal{H}_{fg}$ for executing the parameter optimization. In hidden path selection, g_i^d could be assisted by changing different $\mathbf{H}_{\hat{d}}$ and thus the manifold $\mathcal{H}_{fh}$ could be adjusted to approach the global optimums. Especially, $\mathbf{H}_{\hat{d}}^*$ and $\mathcal{H}_{fh}^*$ represent the optimal $\mathbf{H}_{\hat{d}}$ and $\mathcal{H}_{fh}$ we obtain respectively.

where $\cdot$ represents the matrix product. Importantly, $\partial L/\partial \mathbf{F}^*{}_i$ and $\partial L/\partial \mathbf{W}_i^{*d}$ should be null tensor $\mathbf{0}$ to retain the probability that $\mathbf{F}^*{}_i$ and $\mathbf{W}_i^{*d}$ equal global optimums, which are denoted as $\mathbf{F}^\circ{}_i$ and $\mathbf{W}_i^{\circ d}$, and $\mathcal{H}_f$ contains the global optimums.

Thus, further with the direct functional relationships $\mathbf{W}_i^d = g_i^d(\mathbf{F}_i)$, we require two formulas tenable that

$$\sum_{d<D}\left(\frac{\partial L}{\partial f_d}\cdot\frac{\partial f_d}{\partial \mathbf{F}^*{}_i\odot\mathbf{W}_i^{*d}}\right)\cdot g_i^d(\mathbf{F}_i)=\Lambda_i\cdot G_i\left(\Lambda_{\mathbf{F}^*{}_i}\right)=\mathbf{0}$$
$$\sum_{d<D}\left(\frac{\partial L}{\partial f_d}\cdot\frac{\partial f_d}{\partial \mathbf{F}^*{}_i\odot\mathbf{W}_i^{*d}}\right)\cdot \mathbf{F}^*{}_i=\Lambda_i\cdot\Lambda_{\mathbf{F}^*{}_i}=\mathbf{0} \quad (6)$$

where Λ_i and $\Lambda_{\mathbf{F}^*{}_i}$ are block matrices that list all product factors in the summation on the left side of (6), while function G_i is totally determined by g_i^d. Consequently, (6) illustrates that the column vectors of $\Lambda_{\mathbf{F}^*{}_i}$ and $G_i(\Lambda_{\mathbf{F}^*{}_i})$ should be all within the null space $\mathcal{N}_i$ of Λ_i, which requires numbers of specific points in $\mathcal{N}_i$ to be mapped within $\mathcal{N}_i$ through G_i. In other words, the optimization of $\boldsymbol{\theta}_f$ can be seen within a manifold $\mathcal{H}_{fg}\subset\mathcal{H}_f$, which is narrow down, due to the extra functional relationships $\mathbf{W}_i^d = g_i^d(\mathbf{F}_i)$. Then, the extra constraint illustrated above caused by (6) may not be satisfied by optimal $\boldsymbol{\theta}_f$, which means the narrow $\mathcal{H}_{fg}$ could exclude the global optimums.

Although $\mathcal{N}_i$ is a linear space, nonlinear units such as softmax or sigmoid are always utilized to fix the ranges of $G_i(\Lambda_{\mathbf{F}^*{}_i})$. Thus, as illustrated at the top of Fig. 3, the designed g_i^d, which determines G_i, is hard to meet the required property [i.e., meeting (6)] without the prior knowledge of $\mathcal{N}_i$. Consequently, the narrowed manifold $\mathcal{H}_{fg}$ could exclude mentioned global optimums no matter how elaborate the g_i^d is. The same problem would also exist when $\mathbf{W}_i^{*d}$ has functional relationships with previous layers involved in the calculations of $\mathbf{F}^*{}_i$, because it also requires g_i^d to meet the additional property and all restrictions could only be released by optimizing the parameters $\boldsymbol{\theta}_f$.

C. Hidden Path Selection

To release the restriction of g_i^d caused by direct dependences between $\mathbf{F}_i$ and $\mathbf{W}_i^d$ analyzed in Section III-B, we naturally introduce extra variables $\mathbf{H}_d$ highly independent with $\mathbf{F}_i$ to participate in the calculations in g_i^d, namely $\mathbf{W}_i^d = g_i^d(\mathbf{F}_i, \mathbf{H}_d)$. Serving as the probability mask to indicate the adaptive forward paths for each pixel in $\mathbf{F}_i$, $\mathbf{W}_i^d$ is naturally decided by the properties of different areas in the input image $\mathbf{I}$. Thus, we design the $\mathbf{H}_d$ as the mapping function on $\mathscr{F}$ defined in Section III-A that takes $\mathbf{I}$ as the input. Specifically, we have

$$\mathbf{W}_i^d = g_i^d(\mathbf{F}_i, \mathbf{H}_d),\quad \mathbf{H}_d : \mathscr{F}\rightarrow\mathbb{R}^{s_d} \quad (7)$$

where s_d is a positive integer vector which indicates the shape of $\mathbf{H}_d$. With the help of $\mathbf{H}_d$, the structures of g_i^d could be of no restrictions to some extent, where $\mathbf{H}_d$ can adjust $G_i(\Lambda_{\mathbf{F}^*{}_i})$ to approach $\mathcal{N}_i$ for each $\Lambda_{\mathbf{F}^*{}_i}$ in $\mathcal{N}_i$. Equivalently, the shape of manifold $\mathcal{H}_{fg}$ would be adjusted by $\mathbf{H}_d$ to approach the global optimums. Consequently, the global optimal with features in deep structures considered as variables are more likely to be searched during the optimization of $\boldsymbol{\theta}_f$.

Actually, $\mathbf{H}_d$ can be seen as the hidden variables that are unobservable solely given $\mathbf{F}_i$, which inspires us to imitate the hidden markov chain. Specifically, we design a lightweight mini-branch, which generates $\mathbf{H}_d$ to guide forward paths for each pixel of $\mathbf{F}_i$ in the main branch through (4) and (7), namely hidden path selection.

Algorithm 1 HPS-Net

Input: image $\mathbf{I}$, $\mathscr{L}_\mathrm{d} = \{\mathrm{d_k}\}_{k\in\mathbb{N}_+}$ for the main branch, $\mathscr{H}_\mathrm{d} = \{\mathrm{d_m}\}_{m\in\mathbb{N}_+}$ for the mini-branch
Output: Semantic Segmentation Results

$\mathbf{H}_1 = \mathbf{I}, \mathbf{F}_1 = \mathbf{H}_1$;
for $\bar{\mathrm{d}} = 1; \bar{\mathrm{d}} < \mathrm{D}$ **do**
  $\mathbf{H}_{\bar{\mathrm{d}}+1} = h_{\bar{\mathrm{d}}}(\{\mathbf{H}_\mathrm{i}\}_{\mathrm{i}\in\mathbb{N}_+\le\bar{\mathrm{d}}})$;
for $\mathrm{d} = 1; \mathrm{d} < \mathrm{D}$ **do**
  $\hat{\mathrm{d}} = \min\{\mathrm{d_m}\in\mathscr{H}_\mathrm{d}, \mathrm{d}\le\mathrm{d_m}\}$;
  if $\mathrm{d}\in\mathscr{L}_\mathrm{d}$ **then**
    $\mathbf{F}_{\mathrm{d}+1} = f_\mathrm{d}(\{\mathbf{F}_\mathrm{i}\odot g_\mathrm{i}^\mathrm{d}(\mathbf{F}_\mathrm{i}, \mathbf{H}_{\hat{\mathrm{d}}})\}_{\mathrm{i}\in\mathbb{N}_+\le\mathrm{d}})$;
  else
    $\mathbf{F}_{\mathrm{d}+1} = f_\mathrm{d}(\{\mathbf{F}_\mathrm{i}\}_{\mathrm{i}\in\mathbb{N}_+\le\mathrm{d}})$;
$\mathcal{M}_\mathbf{I} = \mathbf{F}_\mathrm{D}$;
final;
return $\arg\max_{\mathrm{l_k}\in\mathbf{L}}\mathcal{M}_\mathbf{I}(\mathbf{p}),\ \forall\ \mathbf{p}\in\Omega$;

D. Hidden Path Selection Network

As illustrated in Fig. 2, in the proposed HPS-Net, we can implement hidden path selection on existing deep structures (e.g., Resnet-101), which serve as the main branch, based on the analyses in Sections III-B and III-C. An extra mini-branch consisting of the same structure as main branch with all channel numbers reduced to 32 is utilized to generate hidden variables $\mathbf{H}_d$ in (7). Specifically, we utilize a layer-index set $\mathscr{L}_d = \{d_k\}_{k\in\mathbb{N}_+}$ for indicating which layer in the main branch to be applied hidden path selection. Denoting the

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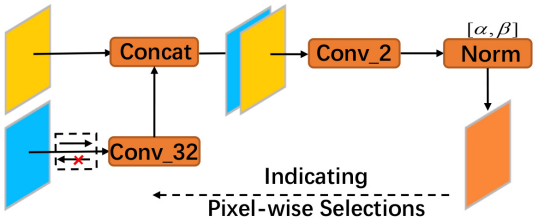


Fig. 4. Proposed HP-Module first utilizes convolutions with channel numbers of 32 and concatenation operations to integrate the feature maps (blue rectangle) from the main branch and hidden variables (yellow rectangle) deriving from mini-branch. Then, the convolution with channel numbers of 2 and normalization functions generate the soft masks (orange rectangle) with values constrained between $[\alpha, \beta]$, which are utilized to indicate the pixelwise path selections for the feature maps in the main branch through multiplication.

basic functions in each layer of mini-branch as $\{h_{\bar{d}}\}_{\bar{d}\in\mathbb{N}_+<D}$, we also utilize a layer-index set $\mathscr{H}_d = \{d_m\}_{m\in\mathbb{N}_+}$ to indicate which feature maps in the mini-branch to serve as hidden variables for hidden path selection. Concretely, we modelize the HPS-Net as

$$\begin{aligned}
&\forall\ \mathbf{H}_1 = \mathbf{I},\quad \bar{d}\in\mathbb{N}_+ < D,\quad \mathbf{H}_{\bar{d}+1} = h_{\bar{d}}\big(\{\mathbf{H}_i\}_{i\in\mathbb{N}_+\le\bar{d}}\big)\\
&\forall\ \mathbf{F}_1 = \mathbf{H}_1,\quad d\in\mathbb{N}_+ < D,\quad \hat{d} = \min\{d_m\in\mathscr{H}_d, d\le d_m\}\\
&\mathbf{F}_{d+1} = \begin{cases} f_d\Big(\big\{\mathbf{F}_i\odot g_i^d(\mathbf{F}_i,\mathbf{H}_{\hat{d}})\big\}_{i\in\mathbb{N}_+\le d}\Big), & \text{if } d\in\mathscr{L}_d\\ f_d\big(\{\mathbf{F}_i\}_{i\in\mathbb{N}_+\le d}\big), & \text{otherwise}\end{cases}\\
&\mathcal{M}_{\mathbf{I}} = \mathbf{F}_D,\quad f(\mathbf{I},\mathbf{p}) = \arg\max_{l_k\in\mathbf{L}}\mathcal{M}_{\mathbf{I}}(\mathbf{p})
\end{aligned}\tag{8}$$

where the output $\mathbf{F}_D$ of main branch finally implements semantic segmentation. The detail process can be found in Algorithm 1.

In the implementation, as illustrated in Fig. 4, we propose HP-Module to generate $\mathbf{W}_i^d = g_i^d(\mathbf{F}_i,\mathbf{H}_{\hat{d}})$, which consists of feature concatenations, convolutions and normalization functions. We have

$$\mathbf{W}_i^d = \big[f_{\text{norm}}^{\alpha,\beta}\big(f_{\text{conv_2}}\big(f_{\text{cat}}\big(f_{\text{conv_32}}(\mathbf{F}_i),\mathbf{H}_{\hat{d}}\big)\big)\big)\big]_i \tag{9}$$

where $f_{\text{conv_2}}$ and $f_{\text{conv_32}}$ are convolutions with 2 and 32 output channels, respectively, where kernel sizes are all set as 3×3. f_{cat} is the concatenation operation. $f_{\text{norm}}^{\alpha,\beta}$ is the normalization function consisting of softmax and clip function, which normalizes the values within $[\alpha,\beta]$. Importantly, $[\,]_i$ indicates the slice alongside the channel dimension according to index i, which is corresponding to the i-th alternative path.

Recalling analyses on the global optimums in Section III-B, we further calculate the gradient flows of $\mathbf{F}_i$ with $\mathbf{W}_i^d = g_i^d(\mathbf{F}_i,\mathbf{H}_{\hat{d}})$ that

$$\frac{\partial L}{\partial \mathbf{F}_i} = \sum_{d<D}\frac{\partial L}{\partial f_d}\cdot\frac{\partial f_d}{\partial \mathbf{F}_i\odot\mathbf{W}_i^d}\cdot\left(\mathbf{W}_i^d + \mathbf{F}_i\cdot\frac{\partial \mathbf{W}_i^d}{\partial \mathbf{F}_i}\right). \tag{10}$$

It is worth noticing that (5) and (10) are not contradicted. Equation (5) calculates the partial derivative of $\mathbf{F}_i$ with features in deep structures (e.g., $\mathbf{W}_i^d$ and $\mathbf{F}_i$) considered as variables, which is utilized to analyze the global optimums in the actual full space and not involved in the calculations among parameter optimization. Equation (10) calculates the gradient flows in our proposed HPS-Net to optimize the parameter $\boldsymbol{\theta}_f$, which can be seen as searching parameters within a manifold $\mathcal{H}_{fh}$ adjusted by $\mathbf{H}_{\hat{d}}$ based on $\mathcal{H}_{fg}$. Moreover, the efficiency of training process would be reduced with $\mathbf{W}_i^d$ being contrary sign to $\mathbf{F}_i\cdot\partial\mathbf{W}_i^d/\partial\mathbf{F}_i$, which is hard to avoid with constantly changing parameters $\boldsymbol{\theta}_f$ during the training process. Consequently, we cutoff the gradient flows from soft masks $\mathbf{W}_i^d$ to feature maps $\mathbf{F}_i$ in the main branch by multiplying null tensor $\mathbf{0}$ with the $\mathbf{F}_i\cdot\partial\mathbf{W}_i^d/\partial\mathbf{F}_i$ on the right side of (10). As we shall see, by controlling gradient flows, the efficiency of training process would be increased.

E. Model Analyses

To hark back to the manifold for parameter optimization, as illustrated in Fig. 3 (bottom), the shape of $\mathcal{H}_{fh}$ could be adjusted by $\mathbf{H}_{\hat{d}}$ based on $\mathcal{H}_{fg}$, where we realize that the high independence between $\mathbf{F}_i$ and $\mathbf{H}_{\hat{d}}$ makes it possible to let $\mathbf{H}_{\hat{d}}$ value freely and adjust $\mathbf{W}_i^d = g_i^d(\mathbf{F}_i,\mathbf{H}_d)$ [i.e., adjust $G_i(\Lambda_{\mathbf{F}^*_i})$] to meet the demanded property. Consequently, the larger range domain of $\mathbf{H}_{\hat{d}}$ in the mini-branch would make $\mathcal{H}_{fh}$ adjusted more freely with the designed formulation of g_i^d, where $\mathcal{H}_{fh}$ is more likely to contain the global optimums with all feature maps considered as variables.

Moreover, the floating range of values in $\mathbf{W}_i^d$ is defined within $[\alpha,\beta]$, which is ensured by applying $f_{\text{norm}}^{\alpha,\beta}$. Recalling that we denote the global optimums $\mathbf{W}_i^d$ as $\mathbf{W}_i^{\circ d}$, we focus on the Taylor expansion with respect to variable $\mathbf{W}_i^d$ that

$$\begin{aligned}
L = {}& L\big(\mathbf{W}_i^{\circ d}\big) + \big(\mathbf{W}_i^d - \mathbf{W}_i^{\circ d}\big)^T\cdot\frac{\partial L}{\partial \mathbf{W}_i^{\circ d}}\\
&+\frac{1}{2}\big(\mathbf{W}_i^d - \mathbf{W}_i^{\circ d}\big)^T\cdot\frac{\partial^2 L}{\partial\big(\mathbf{W}_i^d\big)^2}\cdot\big(\mathbf{W}_i^d - \mathbf{W}_i^{\circ d}\big)\\
&+o\Big(\ \big(\mathbf{W}_i^d - \mathbf{W}_i^{\circ d}\big)^T\cdot\big(\mathbf{W}_i^d - \mathbf{W}_i^{\circ d}\big)\ \Big)
\end{aligned}\tag{11}$$

where $o(\cdot)$ represents the higher order infinitesimal. Since we fix the range of $\mathbf{W}_i^d$ within $[\alpha,\beta]$, the expectation values of $|\mathbf{W}_i^d - \mathbf{W}_i^{\circ d}|$ is $(\beta-\alpha)/2$, which is 0.25 in the implementation. Noticing that $\partial L/\partial\mathbf{W}_i^{\circ d}$ is a null matrix, the second order term in the Taylor expansion would dominate the calculations. Then, with $\partial^2 L/\partial(\mathbf{W}_i^d)^2$ being a positive definite matrix, the loss function L approximates a convex function to some extent with respect to $\mathbf{W}_i^d$. The convexity would make it easier to search the global optimums with freely valuing $\mathbf{W}_i^d$ by adjusting $\mathbf{H}_{\hat{d}}$ if $\mathcal{H}_{fh}$ contains the global optimums.

Consequently, HPS-Net could provide more probability to obtain promising results due to the better manifold for parameter optimization, which can make HPS-Net obtain better pixelwise path selections and better depict pixelwise land-cover distributions during the inference procedure. However, compared with parameter optimization of conventional deep structures without path selections, (10) demonstrates that gradient flows of HPS-Net are considerable small due to the extra multiplication of $\mathbf{W}_i^d$, which might make hidden path selection more effective under small training costs compared with larger training costs.

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F. Loss Function

We utilize fully supervised learning to optimize parameters $\boldsymbol{\theta}_f$ in the proposed HPS-Net. Denoting $\mathcal{G}_{\mathbf{I}}$ as the pixelwise ground truth of $\mathbf{I}$, which indicates the semantic category for each pixel, we have

$$L = \text{Cross_Entropy}(\mathcal{M}_{\mathbf{I}}, \mathcal{G}_{\mathbf{I}}) \tag{12}$$

where the cross entropy function takes the model output $\mathcal{M}_{\mathbf{I}}$ in (8) and ground truth $\mathcal{G}_{\mathbf{I}}$ as the input. Stochastic gradient descent (SGD) is then used in the optimization process of parameter $\boldsymbol{\theta}_f$, through which the mini-branch and main branch can be trained simultaneously.

IV. GID-15 Dataset

As the basis for training and evaluation, a well-annotated benchmark dataset is important to develop remote sensing semantic segmentation algorithms. Although some widely used datasets with several land-cover categories [32], [33], [35], [53] are provided, remote sensing semantic segmentation datasets are hard to give considerations to both the quantities of data samples and sufficiencies of land-cover categories. As a consequence, the increasing demands in real-world applications, such as identifying industrial land and urban residential simultaneously, are difficult to satisfy. For instance, containing abundant data samples that cover dozens of cities in China with 150 images of 6800 $\times$ 7200 pixels, GID-5 [36] ensures the data diversity for deep model training, while the only five involved land-cover categories (i.e., built-up, farmland, forest, meadow and water) are not sufficient for real applications. To propose a better benchmark meeting practical demands, we expand the fine land-cover classification set in [3] by subdividing five land-cover categories in the GID-5 to compose a new GID-15.

Specifically, to further distinguish sub-categories belonging to the same general land-cover categories, we refer to the Chinese Land Use Classification Criteria (GB/T21010-2017) and finally determine 15 concerned categories through a hierarchical category relationships. Concretely, for researching the detailed urban area distributions, we subdivide the built-up into industrial land, urban residential, rural residential, and traffic land. Then, the forest is subdivided into garden land, arbor forest, and shrub land for the researches on vegetation covers, while the meadow is subdivided into natural meadow and artificial meadow. Last but not least, the farmland is subdivided into paddy field, irrigated land and dry cropland for researching agricultural land distributions, while the water is subdivided into river, lake and pond for the water resources researching.

In practice, an expert group checks the original images and annotated category maps in GID-5 simultaneously, through which we recheck the previous annotations and further separate the annotated land-cover regions in GID-5 according to the hierarchical category relationships we build. Similar with GID-5, all the pixels related to obscure land-cover distributions, which are difficult to depict with pixelwise land-cover categories explicitly, are ignored in GID-15. Further due to the wide range areas covered in GID-5, GID-15 can serve as a challenging benchmark containing large quantities of data samples (i.e., 150 images of 6800 $\times$ 7200 pixels with pixelwise annotations) and abundant land-cover categories (i.e., 15 land-cover categories better meeting practical demands) simultaneously. Several samples of GID-15 are displayed in Fig. 5, where we can see the annotation meticulousness. The GID-15 dataset would soon be available at https://captain-whu.github.io/HPS-Net/.

V. Experiments and Analysis

In this section, we evaluate the proposed HPS-Net on both GID-5 and GID-15, where we show the effectiveness of the proposed modules and verify the mathematical analyses in Section III. We first illustrate the evaluation metric in Section V-A, while we further clarify the detail experiment settings in Section V-B. Then, in Section V-C and V-D, we discuss the application of the proposed structures on different classical deep structures [43] and the state-of-the-art pixelwise dynamic algorithm [29], where we explore the merits of proposed hidden path selection. Further in Section V-E, we present detail verifications on the mathematical analyses in Section III and study the effects of each term in the proposed structures through ablation studies and feature visualizations.

A. Evaluation Metrics

To reduce the influence of label imbalance, we follow the principle in [29] that utilize mean intersection over union (mIoU). Denoting the confusion matrix calculated based on $\mathcal{M}_{\mathbf{I}}$ and $\mathcal{G}_{\mathbf{I}}$ as $Q = \{q_{ij}\}$, we have

$$\text{mIoU} = \frac{1}{N}\sum_{i=1}^{N} q_{ii} \Big/ \left[\sum_{j=1}^{N}\left(q_{ij} + q_{ji}\right) - q_{ii}\right] \tag{13}$$

where N is the total number of land-cover categories. With each term on the right-hand side of (13) measuring the identification of one certain category, the average form ensures the equivalence of each category.

B. Experiment Settings

The main deep structures and semantic segmentation algorithms involved in our experiments are as follows.

1) *Resnet-50* [43]: A deep residual structure serving as the main branch in HPS-Net, where 16 skip connections are considered to be applied hidden path selection.
2) *Resnet-101* [43]: A deeper version of residual structure compared with Resnet-50, where 33 skip connections are considered to be applied hidden path selection.
3) *Pspnet* [27]: A semantic segmentation algorithm that utilizes multiscale structures based on poolings of different sizes.
4) *Deeplabv3* [25]: A semantic segmentation algorithm that utilizes multiscale structures based on dilation convolutions.
5) *Deeplabv3** [26]: A semantic segmentation algorithm taking residual structures as the backbone, which integrates the encoder-decoder structures and multiscale structures based on dilation convolutions.

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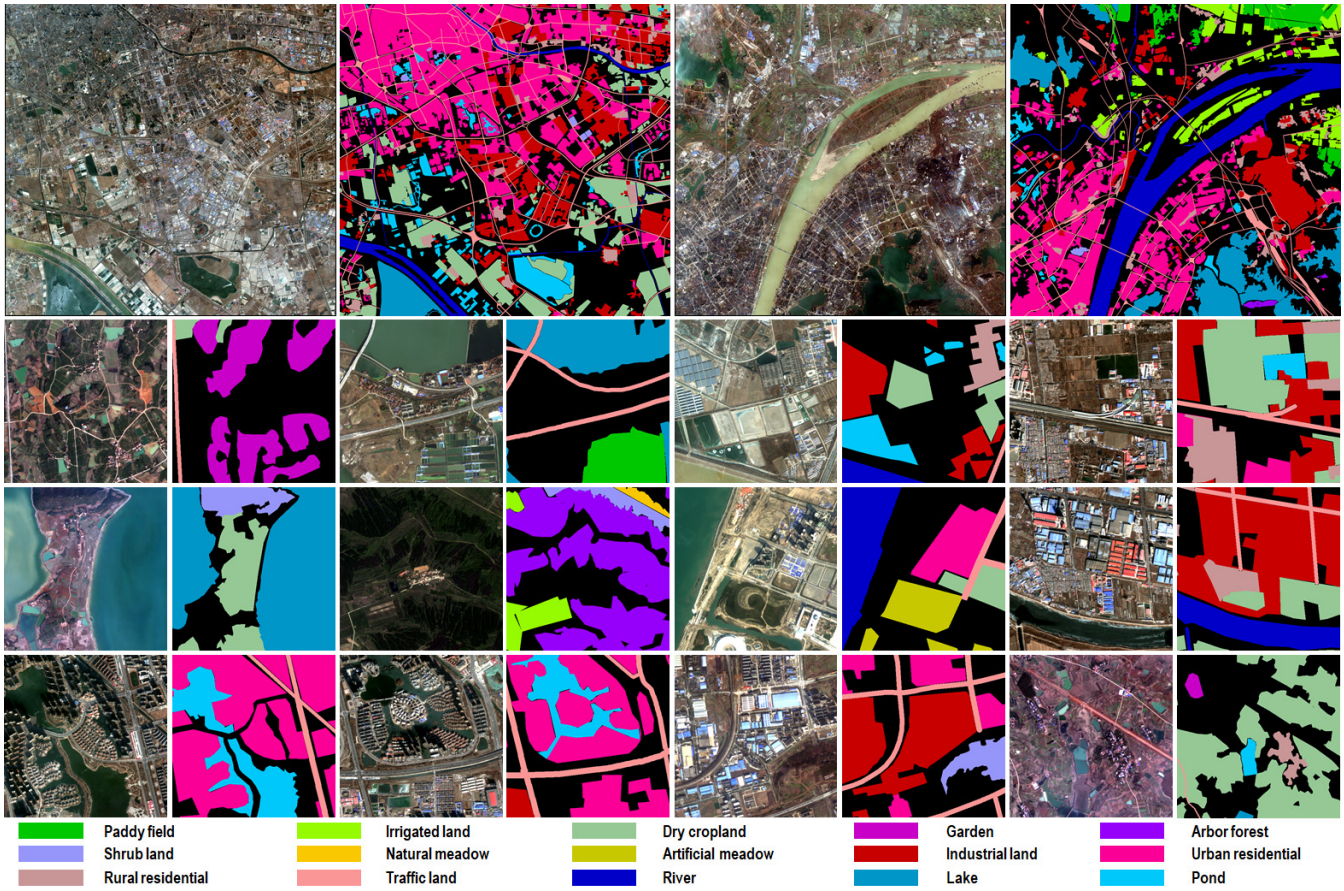


Fig. 5. Data samples taken from our proposed GID-15 dataset. Different colors indicate different land-cover categories. Obscure land covers are blackened in the annotation masks. Several intact data samples are listed at the top and some cropped data patches are listed on the bottom, where we can see the annotation meticulousness with wide geographical areas covered.

6) *GPSNet* [29]: A semantic segmentation algorithm that dynamically selects adaptive forward paths for each pixel.

In this article, we choose residual structures as the main branch in HPS-Net. All the experiments are implemented on a Pascal V100 with 16 GB memory based on Pytorch. All the model training and evaluation are under the same conditions without pretrained parameters and postprocessings. In the training process, SGD is utilized with batch size as 10 and random flip is utilized as the data augmentation. The initial learning rate is set as 0.007 and poly policy is employed with the power of 0.9. We set the momentum as 0.9 and weight decay as 0.0001. In the testing process, we take raw outputs of each model as the results for the evaluation.

As for the hyper-parameters in HPS-Net, we set $\mathscr{L}_d = \{d_k\}_{k\in\mathbb{N}_+}$ as the index of layers containing skip connections, which indicates that total 16 layers in Resnet-50 and 33 layers in Resnet-101 are applied hidden path selection in the main branch. Then, with residual structures divided into several stages according to the pixel resolution, we set $\mathscr{H}_d = \{d_m\}_{m\in\mathbb{N}_+}$ as the index of last layer in each stage, which indicates that total four layers serve as the hidden variables deriving from the mini-branch. Finally, we generally set (α, β) as (0.75, 1.25) in the HP-Module, where the softmax function is multiplied with 2 to avoid the gradient vanishing. Especially, in the first layer of each stage in the main branch, (α, β) are expanded to (0.5, 1.5) due to the additional alternative paths from previous stages.

For all the experiments in both GID-5 and GID-15, we crop the images and corresponding annotation masks into patches of 512 × 512 pixels simultaneously, which are split into two subsets, i.e., totally 25 200 data patches for training and 6300 data patches for testing.

C. Discussions on Hidden Path Selection

For simply discussing the application of hidden path selection on existing deep structures, we first run a toy experiment, where 10% data samples are randomly selected from GID-15 for training and evaluation. Specifically, we apply the hidden path selection on Resnet-101 in Deeplabv3* to compose HPS-Net, which is utilized to compare with the original residual structure. The experimental results with different epochs, i.e., from 10 to 100 epochs at ten intervals, are shown in Fig. 6, where we can see the fold lines marking evaluated mIoU corresponding to each model with the same Resnet-101 backbone.

Specifically, as for the fold line corresponding to Deeplabv3*, we can see that the model encounters a bottleneck around 32.0 in mIoU. However, with only 10% data samples selected, the model training is far from saturation.

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TABLE I

COMPARISONS WITH STATE-OF-THE-ART METHODS TRAINED ON GID-15. WE DENOTE APPLYING THE HIDDEN PATH SELECTION ON DEEPLABV3* AS HPS-NET-101 AND HPS-NET-50, WHILE APPLYING HIDDEN PATH SELECTION ON GPSNET WITH DIFFERENT BACKBONES ARE DENOTED AS HPS-NET-G101 AND HPS-NET-G50. NUMBER 50 AND NUMBER 101 FOLLOWING THE MODEL NAME INDICATE THE RESNET-50 AND RESNET-101 BACKBONES. HPS-NET-V3 INDICATES APPLYING THE HIDDEN PATH SELECTION ON DEEPLABV3. LAND-COVER CATEGORIES LISTED IN THE FIRST ROW ARE ABBREVIATIONS FOLLOWING THE ORDER IN FIG. 5

Methods	*P. field*	*Irr. land*	*Dry cropl.*	*Garden*	*Arb. forest*	*Shr. land*	*Nat. mead.*	*Art. mead.*	*Ind. land*	*Urb. resid.*	*Rur. resid.*	*Traff. land*	*River*	*Lake*	*Pond*	mIoU
RSE-GID [3]	41.2	60.6	19.3	10.8	83.9	2.0	54.4	12.4	45.0	61.1	36.9	2.5	35.8	62.0	13.3	36.1
PSPNet [27]	54.4	74.3	40.1	21.4	85.6	11.9	68.4	30.4	62.9	73.4	61.4	54.9	57.5	68.0	28.7	52.9
Deeplabv3 [25]	51.6	73.8	48.2	16.2	88.2	15.3	65.4	36.3	62.1	72.9	63.0	52.6	**63.8**	**75.3**	**30.7**	54.4
Deeplabv3*-50 [26]	**56.6**	70.2	41.1	21.2	89.2	15.6	64.5	37.9	64.1	73.9	62.5	59.3	60.9	73.5	23.6	54.3
Deeplabv3*-101 [26]	55.7	74.2	44.0	19.5	89.3	12.2	64.0	38.9	64.4	74.4	63.7	57.6	55.2	70.7	27.9	54.1
GPSNet-50 [29]	54.6	72.3	41.1	23.1	90.1	15.6	66.5	38.8	63.6	74.1	63.9	57.1	55.6	71.9	26.7	54.3
GPSNet-101 [29]	53.4	73.4	43.8	16.8	88.8	16.5	66.2	42.0	63.2	73.4	61.2	56.7	52.9	71.5	25.6	53.7
HPS-Net-v3 (Ours)	56.0	73.4	**50.6**	22.6	87.6	15.1	67.3	43.4	64.8	74.6	64.7	57.6	54.5	69.6	23.0	55.0
HPS-Net-50 (Ours)	54.1	73.4	41.9	**24.3**	90.1	13.2	66.7	40.7	**65.2**	**75.0**	**65.2**	**60.8**	57.1	71.4	26.4	55.0
HPS-Net-101 (Ours)	55.2	74.7	47.6	20.8	88.7	12.7	63.9	41.3	64.7	74.6	63.1	59.5	58.9	71.5	25.8	54.9
HPS-Net-G50 (Ours)	56.1	74.0	47.0	20.4	91.0	**17.3**	68.8	44.0	63.9	74.7	**65.2**	59.7	52.1	71.5	26.5	**55.5**
HPS-Net-G101 (Ours)	53.2	**75.6**	47.3	22.1	**91.1**	17.0	**69.5**	**45.2**	63.3	74.0	64.4	59.0	55.6	72.4	23.4	**55.5**

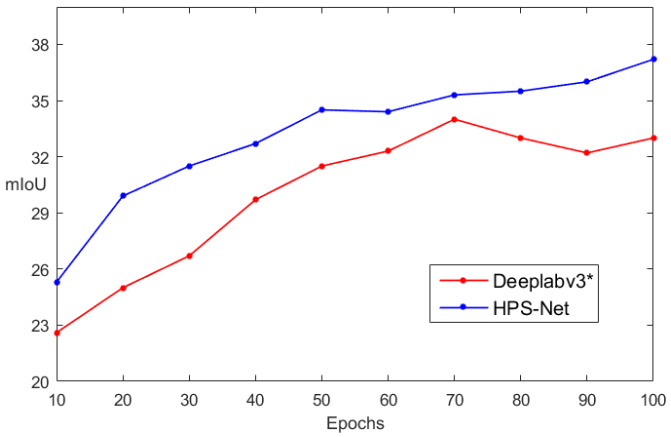


Fig. 6. Toy experiments involving in randomly selected 10% data samples. We train the Deeplabv3* and HPS-Net, where hidden path selection is applied on Resnet-101 in Deeplabv3*, for different training epochs. Larger mIoU of HPS-Net indicates the merits of hidden path selection and the increasing tendency of fold line corresponding to HPS-Net implies the better shape of high dimension manifold, which makes global optimums more accessible.

Thus, this bottleneck indicates that the parameter optimization in Deeplabv3* on the high dimension manifold hovers at the same level, which implies the defects of aforementioned manifold. Then, for the fold line corresponding to HPS-Net, the model performance is constantly improved with increasing training epochs, which implies the better shape of manifold for parameter optimization in HPS-Net to make global optimums more accessible as demonstrated by the mathematical analyses in Section III-E. As for the comparison of twofold lines, HPS-Net outperforms Deeplabv3* by average 3.2 in mIoU, which indicates the improvement attributed to applying hidden path selection on the basic residual structure.

D. Comparison With the State-of-the-Art

To compare with state-of-the-art algorithms, we list the IoU for each land-cover category and mIoU for overall evaluation in Table I, where all models are trained on GID-15 with 20 epochs from scratch. We apply the hidden path selection on Deeplabv3* and GPSNet to compose the HPS-Net. We denote applying the hidden path selection on Deeplabv3* as HPS-Net-50 and HPS-Net-101, while applying hidden path selection on GPSNet with different backbones are denoted as HPS-Net-G50 and HPS-Net-G101. Number 50 and number 101 following the model name indicate the Resnet-50 and Resnet-101 backbones. As we can see, HPS-Net tends to achieve better results on the majority of land-cover categories, while the categories related to water cannot be depicted that well. Furthermore we can conclude from Table I that HPS-Net-G50 and HPS-Net-G101 achieve the best results, i.e., 55.5 in mIoU.

Specifically, HPS-Net-50 outperforms Deeplabv3*-50 by 0.7 in mIoU and HPS-Net-101 raises 0.8 improvement in mIoU over Deeplabv3*-101, while HPS-Net-G50 outperforms GPSNet-50 by 1.2 in mIoU and HPS-Net-G101 raises 1.8 improvement in mIoU over GPSNet-101. The model comparisons listed above illustrate that the applied hidden path selection can stably improve the performance of existing deep structures. Besides, we can check that GPSNet-50 and GPSNet-101 achieve comparative results with Deeplabv3*-50 and Deeplabv3*-101, which indicates that the proposed hidden path selection can build better pixelwise dynamic structures than the gate path selection proposed in the state-of-the-art algorithm GPSNet. However, together with the application of hidden path selection, gate path selection is able to raise more stable improvements according to the comparison between HPS-Net-50 and HPS-Net-G50 or HPS-Net-101 and HPS-Net-G101. We display the semantic segmentation results in Fig. 7, where we can also see the superiorities of the proposed HPS-Net.

For the experiments on GID-5, we recover the five land-cover categories in GID-5 from the data samples involved in experiments mentioned above according to the hierarchical category relationships in Section IV, based on which all models are also trained for 20 epochs from scratch.

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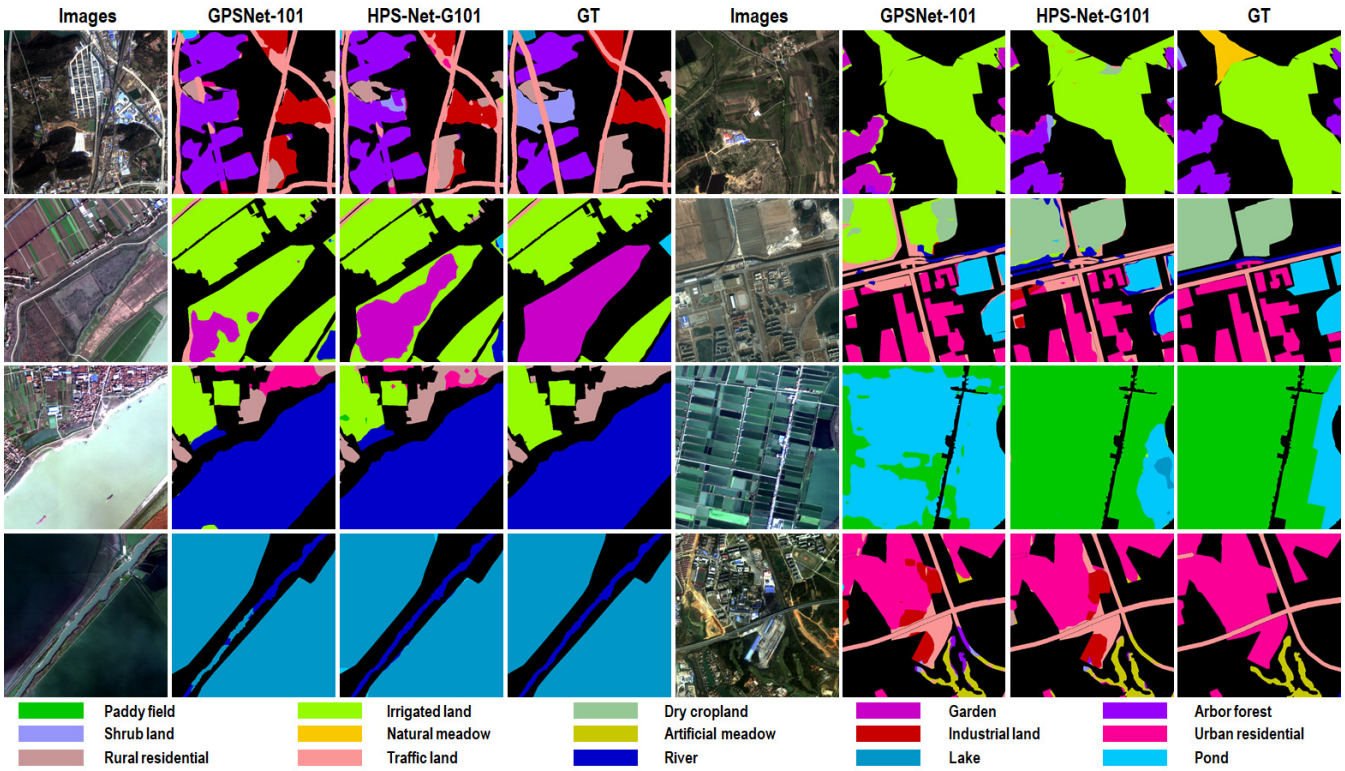


Fig. 7. Examples of comparing our HPS-Net with the state-of-the-art pixelwise dynamic method GPSNet on GID-15 with Resnet-101 as the backbone. Our proposed HPS-Net can better depict pixelwise land-cover distributions. Blackened pixels are the ignored regions in the annotation masks, which are also ignored in the semantic segmentation results.

The experimental results are summarized in Table II, where we can also see that HPS-Net obtains better results. Moreover, as illustrated in Tables I and II, the patch-based method proposed in [3] could not perform as well as methods which parse semantic labels for each pixel directly. However, it is worth noticing that, the gaps between HPS-Net and other algorithms are smaller than that in the experiment results on GID-15, which demonstrates that GID-15 can better evaluate different semantic segmentation algorithms. We also illustrate the semantic segmentation results in Fig. 8 to show the merits of applying hidden path selection on the basic residual structures.

TABLE II
COMPARISONS WITH STATE-OF-THE-ART METHODS ON GID-5

Methods	*Farmland*	*Forest*	*Meadow*	*Built-up*	*Water*	mIoU
RSE-GID [3]	77.5	81.8	39.9	73.2	86.7	71.8
PSPNet [27]	89.7	87.2	53.7	93.8	**92.1**	83.3
Deeplabv3 [25]	89.5	88.2	51.6	93.7	92.0	83.0
Deeplabv3*-50 [26]	**90.1**	87.5	53.2	94.1	91.8	83.3
Deeplabv3*-101 [26]	89.4	86.9	54.3	93.7	91.1	83.1
GPSNet-50 [29]	90.0	88.0	55.6	93.7	92.0	83.9
GPSNet-101 [29]	88.6	86.7	51.9	91.6	91.0	82.0
HPS-Net-v3 (Ours)	90.0	**88.4**	55.7	94.0	92.0	84.0
HPS-Net-50 (Ours)	90.0	86.7	57.0	94.4	91.2	83.8
HPS-Net-101 (Ours)	**90.1**	87.6	57.1	**94.5**	92.0	**84.3**
HPS-Net-G50 (Ours)	**90.1**	87.0	**57.2**	93.9	91.4	83.9
HPS-Net-G101 (Ours)	90.0	88.2	55.8	94.0	91.9	84.0

E. Ablation Study and Feature Visualization

To study the effects of each term in the proposed structures, we design several ablation studies in this section. In HPS-Net-50-ps and HPS-Net-101-ps, we follow [24] to share forward path selections within each image, where we can see that pixelwise path selections, i.e., HPS-Net-50 and HPS-Net-101, obtain better results. Then, we fix the hidden variables in HPS-Net, denoted as HPS-Net-50-fh and HPS-Net-101-fh, to study the influence of direct relationships between $\mathbf{W}_i^d$ and $\mathbf{F}_i$ analyzed in Section III-B. Besides, we further keep intact gradient flows in the HP-Module, denoted as HPS-Net-50-ig and HPS-Net-101-ig, to study the gradient analyses in Section III-D.

As illustrated in Table III, HPS-Net-50-ps and HPS-Net-101-ps are inferior to HPS-Net-50 and HPS-Net-101 by 0.5 and 2.2 in mIoU, respectively, which indicates the superiority of pixelwise dynamic structures. Then, HPS-Net-50 and HPS-Net-101 outperform HPS-Net-50-fh and HPS-Net-101-fh by 2.4 and 2.8 in mIoU respectively, which verifies the deleteriousness of direct relationships between $\mathbf{W}_i^d$ and $\mathbf{F}_i$ analyzed in Section III-B. Finally, HPS-Net-50-ig and HPS-Net-101-ig achieve lower results by 1.4 and 1.8 in mIoU than HPS-Net-50 and HPS-Net-101 respectively, which verifies the inefficiency of model training with intact gradient flows analyzed in Section III-D.

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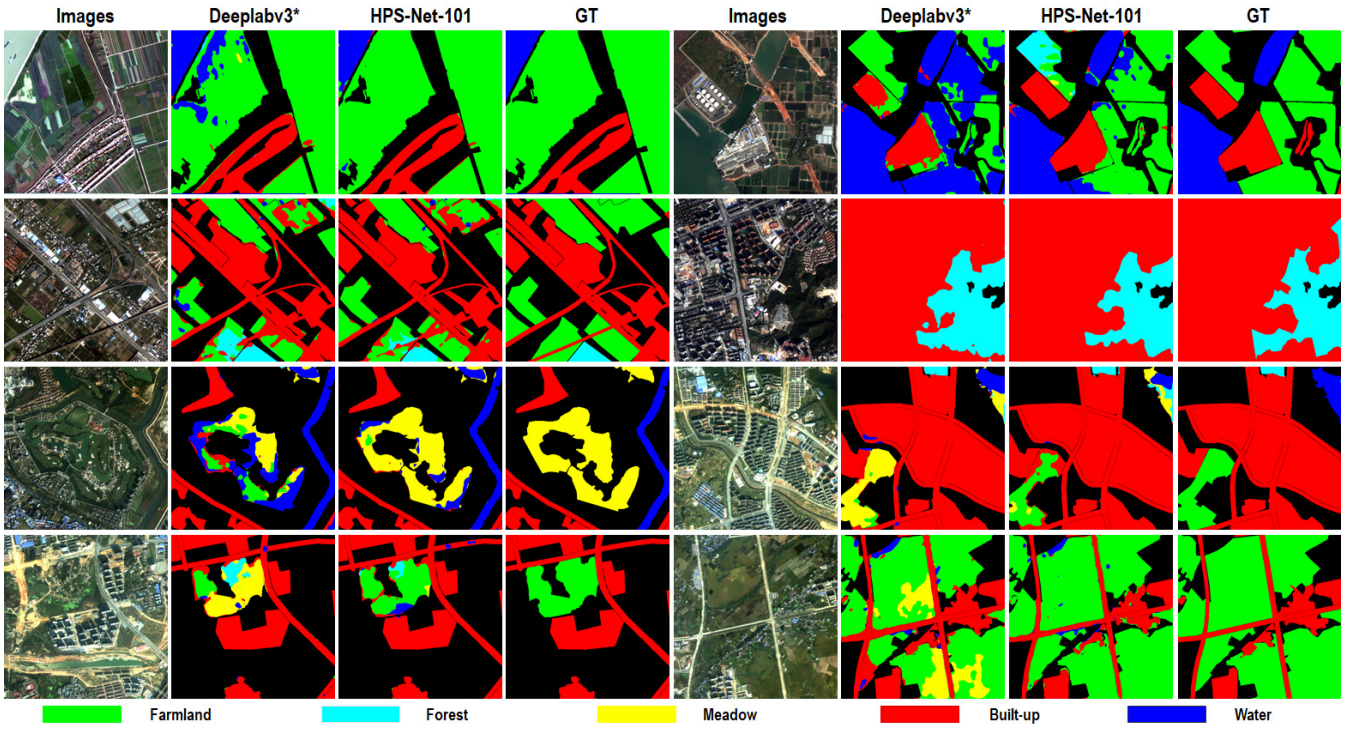


Fig. 8. Examples of comparing our HPS-Net with the basic residual structures Deeplabv3* on GID-5 with Resnet-101 as the backbone, which display the merits of applying the proposed hidden path selection on Deeplabv3* (i.e., HPS-Net-101).

TABLE III

ABLATION STUDIES ON GID-15. HPS-NET-50-PS AND HPS-NET-101-PS REPRESENT ALL PIXELS SHARING THE SAME DYNAMIC FACTOR. HPS-NET-50-IG AND HPS-NET-101-IG REPRESENT INTACT GRADIENT FLOWS IN HP-MODULE. HPS-NET-50-FH AND HPS-NET-101-FH REPRESENT FIXING THE HIDDEN VARIABLES TO NULL TENSORS IN HP-MODULE

Methods	*P. field*	*Irr. land*	*Dry cropl.*	*Garden*	*Arb. forest*	*Shr. land*	*Nat. mead.*	*Art. mead.*	*Ind. land*	*Urb. resid.*	*Rur. resid.*	*Traff. land*	*River*	*Lake*	*Pond*	mIoU
HPS-Net-50-ps	52.5	72.9	42.0	24.4	88.2	15.3	64.1	40.7	64.8	74.9	64.5	60.2	59.5	69.6	24.3	54.5
HPS-Net-101-ps	54.4	72.8	46.3	19.4	89.9	14.4	61.5	33.6	59.4	73.6	61.3	55.4	50.7	68.2	29.2	52.7
HPS-Net-50-fh	52.0	71.8	37.9	22.8	85.7	12.8	63.1	39.2	63.9	74.5	63.8	59.9	53.8	65.5	22.3	52.6
HPS-Net-101-fh	57.1	72.4	40.6	20.6	87.5	12.5	62.9	34.7	62.8	73.6	63.5	57.6	48.9	64.6	22.7	52.1
HPS-Net-50-ig	56.0	71.2	39.9	21.4	86.2	13.0	65.0	38.9	63.6	74.0	63.6	58.4	57.4	70.3	25.2	53.6
HPS-Net-101-ig	55.9	72.2	38.0	20.6	88.9	14.5	62.3	41.8	64.5	74.1	63.0	58.5	47.9	68.3	25.7	53.1
HPS-Net-50	54.1	73.4	41.9	24.3	90.1	13.2	66.7	40.7	65.2	75.0	65.2	60.8	57.1	71.4	26.4	55.0
HPS-Net-101	55.2	74.7	47.6	20.8	88.7	12.7	63.9	41.3	64.7	74.6	63.1	59.5	58.9	71.5	25.8	54.9

For illustrating the depiction of pixelwise land-cover distributions by HPS-Net during the inference procedure, we visualize the selected paths of several data samples in Fig. 9. As illustrated in the first row of Fig. 9, in contrast to [24], the proposed HPS-Net considers in the scale diversity among land covers within one image and then selects different forward paths for different image regions. Specifically, HPS-Net emphasizes paths with more complex computations, i.e., the first alternative path, among rural residential located at the top area of the image, which alleviates the mis-predictions. And the comparisons with pixelwise path selections in existing algorithm [29], i.e., GPSNet, show the merits of proposed modules and validate the mathematical analyses in Section III-B. Moreover, as shown in the second row, the combination of path selections from first stage to third stage ensures different forward paths for irrigated land at left-top and dry cropland at right-bottom, which makes efforts to distinguish these two land-cover categories in the same image and also alleviate the mis-prediction. As we can see, the proposed HPS-Net can better depict pixelwise land-cover distributions during the inference procedure. Moreover, we display the masks corresponding to path selection in HPS-Net-G101 and GPSNet-101 when the methods are applied on the GID-15 task and GID-5 task in Fig. 10. We can conclude from the Fig. 10 that HPS-Net would adaptively generate more complex path selections for more complicated task.

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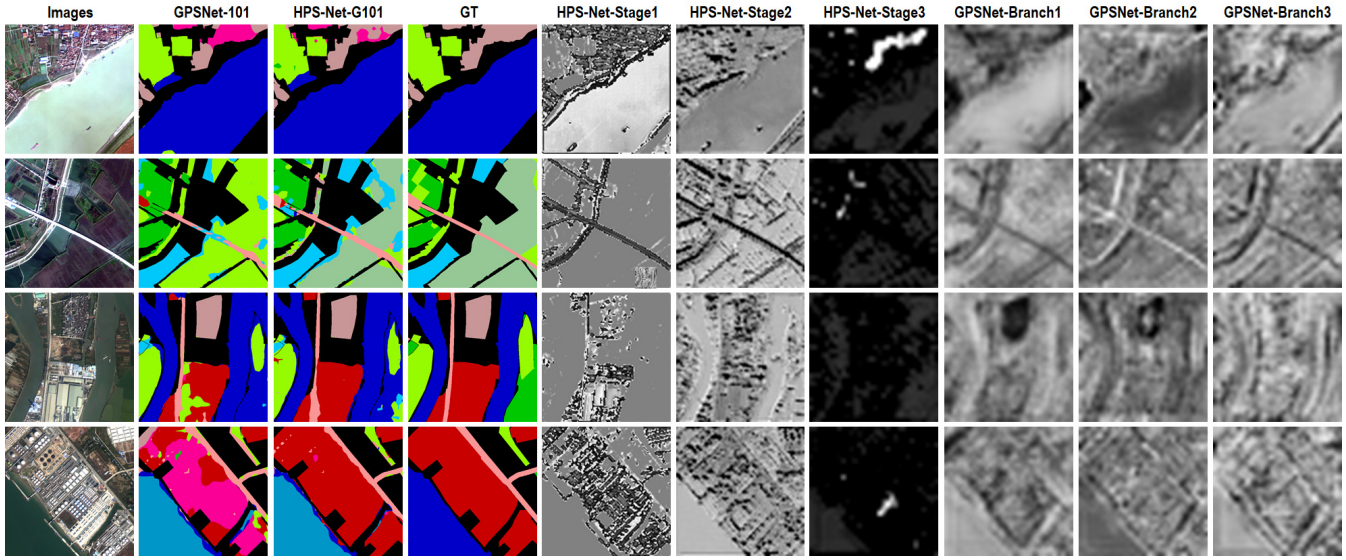


Fig. 9. Visualization of the masks corresponding to path selection in HPS-Net-G101 and GPSNet-101. We visualize the path selection for the first layer of first stage to third stage in HPS-Net-G101, while the first mask of first branch to third branch in GPSNet are listed too. Brighter grayscale values in these masks indicate larger factors corresponding to the first alternative path, where the proposed HPS-Net can better depict pixelwise land-cover distributions through diverse forward path selections for different pixels compared with GPSNet.

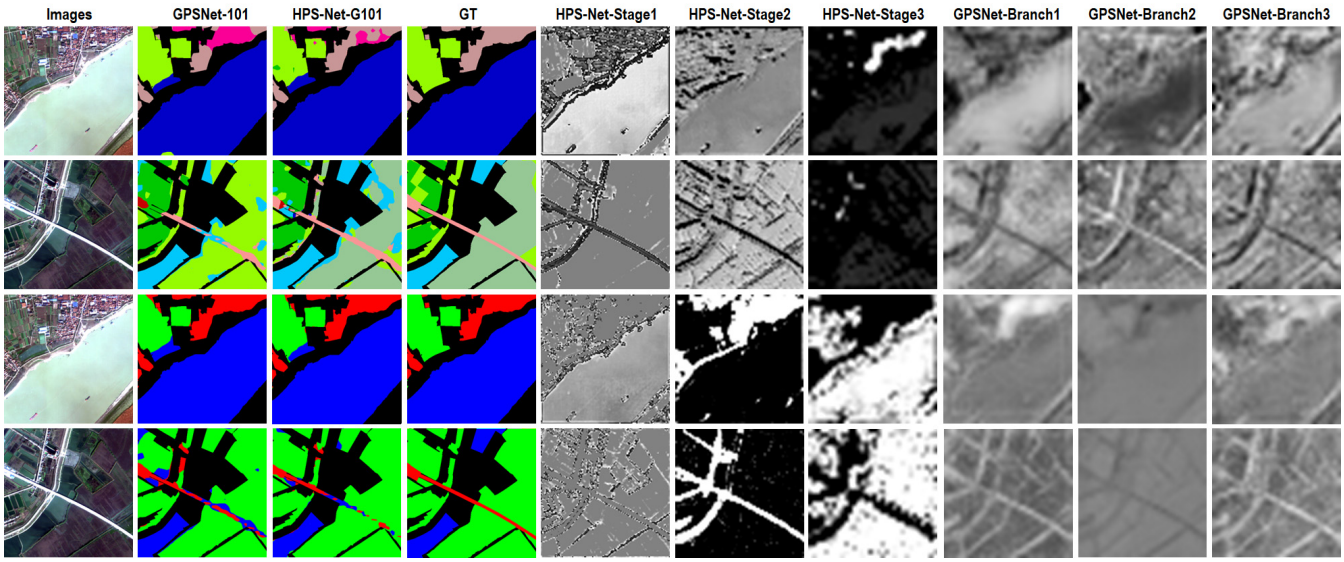


Fig. 10. Visualization of the masks corresponding to path selection in HPS-Net-G101 and GPSNet-101 when the methods are applied to the same images with different number of labels. First two rows are corresponding to GID-15 task and the other rows are corresponding to GID-5 task. Compared with the GID-15 task, the masks corresponding to path selection in GID-5 task are more concise, which indicates that HPS-Net would adaptively generate more complex path selections for more complicated task.

Then, to explore the efficiency of proposed HPS-Net, we list the computation costs calculated based on patches of GID-15 for each model in Table IV. As we can see, pixelwise dynamic structures cause 5.7G extra FLOPs and hidden variables result in 0.3G extra FLOPs with Resnet-50 as the backbone, while the corresponding extra FLOPs are 6.0G and 0.4G, respectively, with Resnet-101 as the backbone. Compared with the FLOPs of Deeplabv3*, HPS-Net results in less than 10% extra computation costs, which indicates that HPS-Net gives the consideration to both effectiveness and efficiency.

Finally, we implement the additional experiments on PASCAL VOC [56] and Cityscapes [57] to briefly compare the proposed methods with [24] and other computer vision methods. We follow the original experimental settings of each backbone for HPS-Net and utilize stepwise strategy to train the hidden path selection taking the multibranch in each backbone as alternative paths, where the sigmoid function is utilized as the normalization function. In Table V, we display the application results of hidden path selection on Resnet-101, where we can see that hidden path selection can improve the model performance compared with Deeplabv3* and HPS-Net performs

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TABLE IV

ABLATION STUDIES ON THE COMPUTATIONAL EFFICIENCY OF HIDDEN PATH SELECTION. HPS-NET-W/O-H MEANS REMOVING HIDDEN VARIABLES. ALL THE NUMBERS REPRESENT FLOPS (G) CALCULATED BASED ON PATCHES OF GID-15 FOR EACH MODEL

Backbones	Deeplabv3*	HPS-Net-w/o-h	HPS-Net
Resnet-50	69.2	74.9	75.2
Resnet-101	88.6	94.6	95.0

TABLE V

COMPARISON EXPERIMENTS ON PASCAL VOC VAL SET

Methods	Backbone	mIoU
Deeplabv3[25]	MobileNet[54]	75.3
Deeplabv3[25]	MobileNetV2[55]	75.7
Auto-Deeplab[28]	Searched-F20[28]	78.3
Deeplabv3*[26]	Resnet-101[43]	78.9
Dynamic[24]	Layer16[24]	78.6
Dynamic[24]	Layer33[24]	78.9
HPS-Net(Ours)	Resnet-101[43]	**79.1**

TABLE VI

COMPARISON EXPERIMENTS ON CITYSCAPES VAL SET

Methods	Backbone	mIoU
Deeplabv3[25]	Resnet-101[43]	78.5
Deeplabv3*[26]	Xception-71[58]	79.6
PSPNet[27]	Resnet-101[43]	79.7
Autp-Deeplab[28]	Searched-F20[28]	79.7
Autp-Deeplab[28]	Searched-F48[28]	80.3
Dynamic[24]	Layer33[24]	79.2
OCRNet[59]	HRNet[60]	81.1
HPS-Net(Ours)	Layer33[24]	79.4
HPS-Net(Ours)	HRNet[60]	81.3
HPS-Net(Ours)	ViT[61]	**84.9**

better than [24]. Then in Table VI, we show the experimental results of HPS-Net with different backbones, where we can conclude that hidden path selection can improve the model performance with different backbones and HPS-Net obtains the best result when not utilizing coarse set of Cityscapes compared with other comparison methods.

In summary, we can see in Tables I, II, V, and VI that with the application of hidden path selection all the models involved in experiments obtain the performance improvements. Also, the comparisons between the HPS-Net and GPSNet in Tables I and II and Fig. 9 show that the path selection method in HPS-Net is superior to the one in GPSNet, which validates the mathematical analyses in Section III-B. Further with the conclusion from Table IV, it can be found out that the proposed method can raise model improvements with minimal costs. Actually, the structures of deep networks can decide the shape of the manifold for parameter optimization, which influence the model performance to a large extent. How to make quantitative analyses of the relationship between structures of deep networks and manifold for parameter optimization is a meaningful topic for the future work.

VI. CONCLUSION

In this article, we propose a HPS-Net for semantic segmentation in remote sensing images to select adaptive forward paths for every pixel guided by the mathematical analyses in terms of the parameter optimization. The inherent problem about inaccessible global optimums is tackled with the help of hidden variables deriving from an extra mini-branch in HPS-Net. For the better training and evaluation, we expand the land-cover classification dataset, i.e., GID-5, into 15 land-cover categories and propose the new GID-15 dataset, which provides more abundant land-cover categories and better satisfies the real-world scenarios. The experimental results on both GID-5 and GID-15 show that the proposed module can stably improve the performances of existing deep structures.

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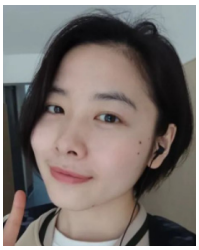
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